Bright Coffee Shop Sales Analysis: Jan-June 2023

Driving Revenue Growth through Data-Driven Insights

Table of Contents

1. Overview
2. Process
3. Summary
4. Tools Used

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1. Overview

1.1 Project Background

Bright Coffee Shop wanted to move from raw transaction logs to strategic decisions. As a Junior Data Analyst, I was tasked with analyzing 6 months of sales data (Jan-June 2023) to drive revenue growth and improve product performance.

1.2 Business Objectives

Our mission was clear: **grow revenue and improve product performance.**

1.3 Key Business Questions

This analysis was designed to answer four critical CEO-level questions:

1. Which products generate the most revenue?
2. What are peak sales times?
3. What sales trends exist across products and time?
4. Which products underperform and how can we improve them?

1.4 Dataset

- **Source:** Bright Coffee Shop sales dataset from Excel
- **Volume:** 149,116 records
- **Period:** January - June 2023
- **Key Fields:** transaction date/time, store location, product_id, product_detail, quantity, unit price, total amount

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2. Process

I followed a structured 6-step analytics workflow to turn raw data into CEO-ready insights.

Workflow: `Business Questions -> Data Source & Ingestion -> ETL & Processing -> SQL Analysis -> Excel & Dashboards -> CEO Decision Support`

01 - SOURCE

Ingested raw sales data from Excel file containing 149,116 records.

02 - VALIDATE & CONVERT

- Performed data validation checks for missing values and duplicates.
- Converted Excel data to CSV format for Databricks.
- Identified formatting issues like decimal commas.

03 - CLEAN & PREPARE IN DATABRICKS

ETL operations:

- Fixed decimal commas and inconsistent formatting
- Cast numeric fields (price, quantity) to correct data types

- Created time buckets (hourly, daily, monthly) to analyze peak sales times
- Calculated `total Amount` = quantity * unit price

04 - SQL GROUPING & AGGREGATION

Used SQL for deep analytical grouping:

```
SELECT product_id, time_bucket, SUM(total_amount) AS total_revenue
FROM sales_clean
GROUP BY product_id, time_bucket, other_dimensions;
```

05 - EXPORT PROCESSED DATA

Exported cleaned and aggregated tables to Excel for visualization.

06 - ANALYSE & VISUALISE

Built dynamic Excel dashboards with Pivot Tables, Slicers, and Charts for Sales Overview, Total Revenue, Top Products.

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3. Summary

3.1 Key Sales Findings

Metric	Insight
$698,800	Total Revenue from Jan-June 2023
66.7%	Coffee (38.6%) and Tea (28.1%) together drive 2/3 of revenue
89%	Coffee, Tea, Bakery, and Drinking Chocolate combined = 89% of revenue

Metric	Insight
Top Locations	"Others" leads overall revenue; Location 6 is top named location
Top Products	Product Details 1 and 2 drive highest transaction quantities, followed by 3

3.2 CEO Recommendations for Growth

01. Run targeted campaigns during slow time slots
02. Stock more of top sellers like Coffee and Tea
03. Promote underperforming items to balance the portfolio
04. Automate daily sales reporting for faster decisions
05. Track performance across store locations for better insights

3.3 From Raw Data to Strategic Decisions

Raw transactions -> Aggregated with SQL -> Visualized in dynamic Excel dashboards -> Translated into business decisions -> Decisions support CEO goals and drive growth.

Final Outcome: Data-driven insights powering better decisions and sustainable revenue growth.

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4. Tools Used

Category	Tools
Data Source	Microsoft Excel (149,116 records)
Processing & ETL	Databricks, CSV
Analysis	SQL (Grouping, Aggregation)

Category	Tools
Visualization	Excel (Pivot Tables, Dashboards)
Version Control	Git & GitHub

Project Structure

```
/bright-coffee-shop-sales-analysis
├── /data
├── /databricks
├── /excel_dashboard
└── README.md
```

Author

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